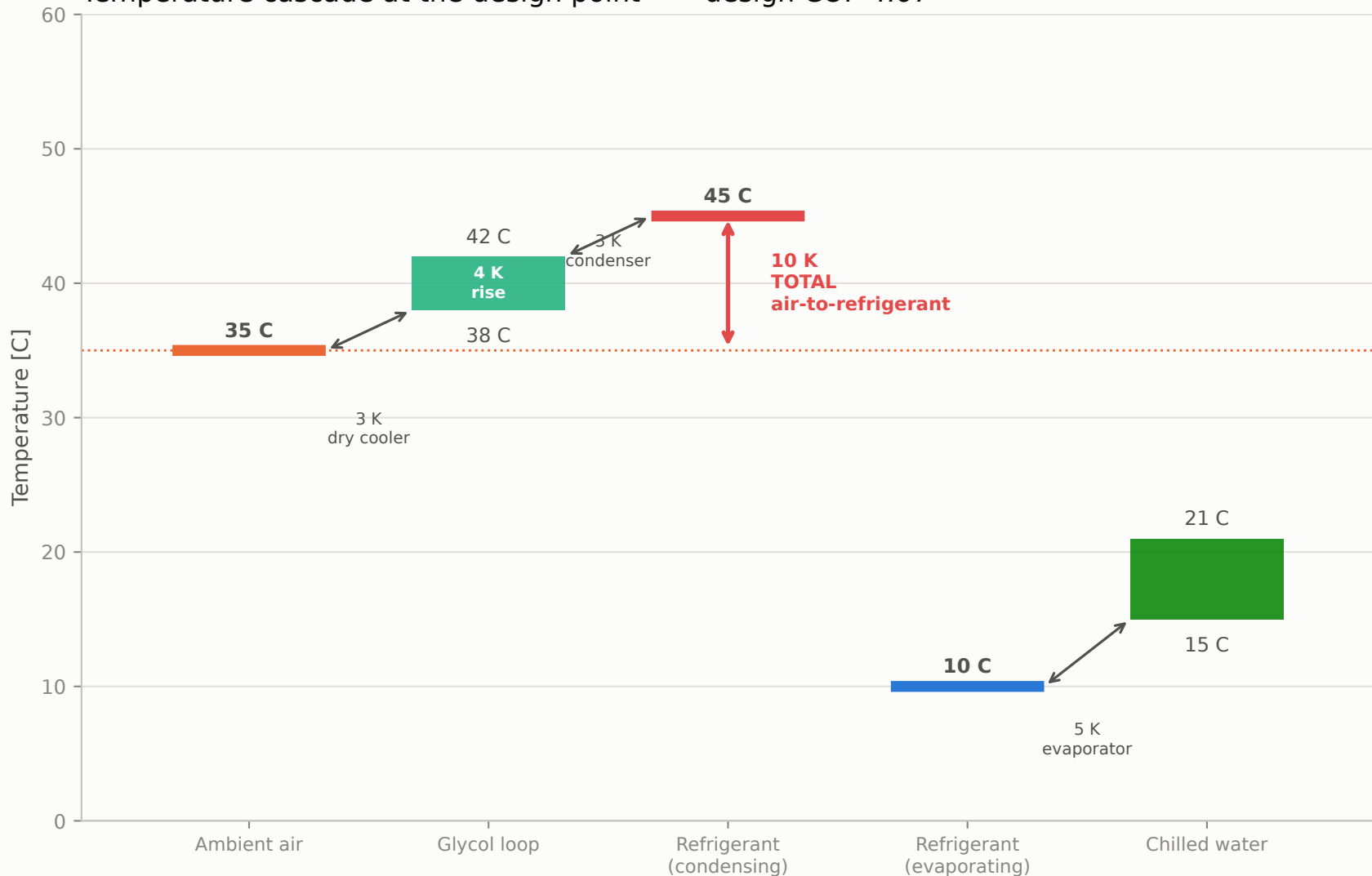


Temperature cascade at the design point --- design COP 4.67



The intermediate glycol loop stacks three approaches in series between ambient air and the refrigerant --- 3 K across the dry cooler, a 4 K glycol rise, 3 K across the condenser. Their sum is the 10 K that sets t_c , and t_c sets the COP. This is the price of keeping an A3 refrigerant out of the data hall, and the reason the dry cooler is sized as generously as it is.